Inter-Annotator Reliability Analysis — Gender Occupation Bias Stream

To assess inter-annotator consistency within the gender occupation bias evaluations, an Intra-Class Correlation Coefficient (ICC(2,1)) analysis was conducted using a two-way random, absolute-agreement model. The resulting ICC value was 0.97, with a 95% confidence interval ranging from 0.955 to 0.982, indicating strong inter-rater reliability. According to conventional thresholds (Koo & Li, 2016), this level of agreement reflects *excellent consistency* among annotators, suggesting that observed rating variance primarily reflects true differences in stimulus bias rather than random or subjective discrepancies across raters.

Complementary analyses using pairwise correlation matrices further support this conclusion. Most Pearson correlation coefficients between raters fell within the 0.6–0.8 range, demonstrating moderate to strong linear alignment in scoring patterns. A small number of lower or missing correlations likely reflect limited item overlap or occasional interpretive divergence but do not undermine the overall robustness of agreement.

To examine ordinal consistency, Spearman's rank (ρ) and Kendall's tau (τ) correlations were also computed. Both rank-based measures showed generally high correspondence ($\rho \approx 0.65$–0.85, $\tau \approx 0.55$–0.78) across raters, indicating strong alignment in the *relative ordering* of perceived bias severity. Together, these results confirm that raters exhibited highly similar evaluative patterns, both in absolute scores and in the relative ranking of biased responses.

Overall, the gender occupation bias stream demonstrates excellent inter-annotator reliability, suggesting that the rubric was well-calibrated and consistently applied across raters.